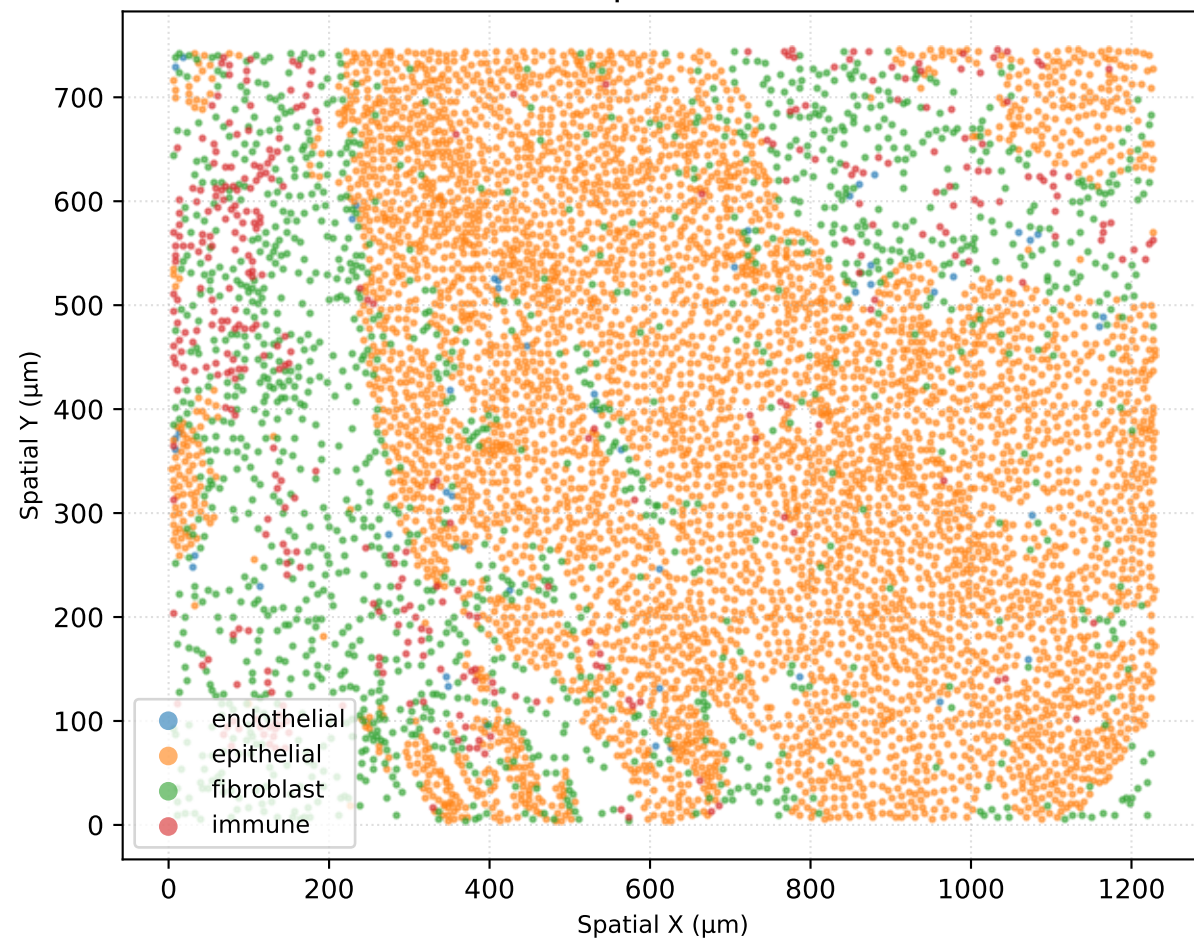
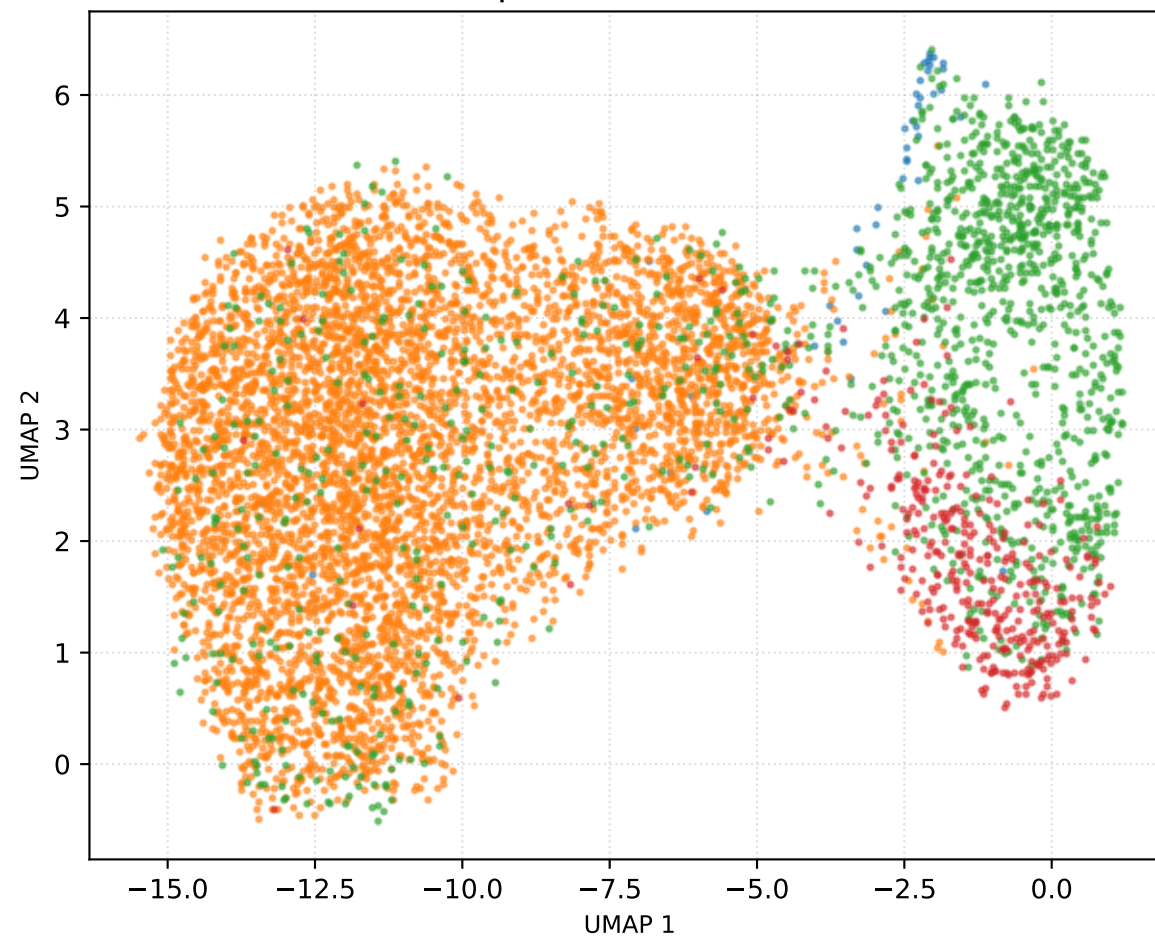


Cell-Type Structure Preservation: Feature Comparison across Paradigms + Spatial Control
[scGPT Zero-Shot: sil=0.1359 (ratio 0.92) | Spatial (x,y) Control: sil=0.0213 (ratio 0.14) | PCA-50: sil=0.1476 (ratio 1.00) | BiLSTM-50: sil=0.4287 (ratio 2.91)]

Spatial Position Alone (x,y Control)
Silhouette: 0.0213 | Ratio vs PCA-50: 0.1445



PCA-50 Baseline (Raw Expression)
Silhouette: 0.1476 | Ratio vs PCA-50: 1.00 (Reference)



BiLSTM-50 Spatial Neighbor Extractor
Silhouette: 0.4287 | Ratio vs PCA-50: 2.9051

